

AI Code Review Report

Generated: 2026-07-08 10:17:59

Overall Quality Score

56/100

Score Breakdown

Metric	Score
Readability	40/100
Security	50/100
Performance	92/100
Maintainability	50/100
Documentation	50/100

Complexity: Medium

Issues Found

Bugs

- Line 59: Potential infinite loop detected (*high*)
- Line 161: Division by zero detected (*critical*)

Security Issues

- Line 15: Hardcoded credential found (*high*)
- Line 51: Potential Command Injection vulnerability (*critical*)
- Line 76: eval/exec usage detected (*critical*)
- Line 81: Pickle is insecure for untrusted data (*high*)
- Line 86: Weak cryptographic algorithm "md5" used (*high*)

Performance Issues

- Line 109: Using range(len()) - consider using enumerate() (*low*)

Style Issues

- Line 30: Missing docstring (*medium*)
- Line 31: Missing docstring (*medium*)
- Line 42: Missing docstring (*medium*)
- Line 50: Missing docstring (*medium*)
- Line 54: Missing docstring (*medium*)
- Line 58: Missing docstring (*medium*)
- Line 63: Magic number 50 found (*low*)
- Line 67: Missing docstring (*medium*)
- Line 68: Magic number 42 found (*low*)
- Line 75: Missing docstring (*medium*)
- Line 79: Missing docstring (*medium*)
- Line 85: Missing docstring (*medium*)
- Line 90: Missing docstring (*medium*)
- Line 100: Missing docstring (*medium*)
- Line 107: Missing docstring (*medium*)
- Line 115: Line is 129 characters (max 100) (*low*)
- Line 115: Missing docstring (*medium*)
- Line 119: Magic number 50 found (*low*)
- Line 121: Magic number 75 found (*low*)
- Line 126: Missing docstring (*medium*)
- Line 132: Missing docstring (*medium*)
- Line 139: Missing docstring (*medium*)

- Line 143: Missing docstring (*medium*)
- Line 147: Line is 189 characters (max 100) (*low*)
- Line 150: Missing docstring (*medium*)
- Line 152: Magic number 20 found (*low*)
- Line 153: Magic number 30 found (*low*)
- Line 154: Magic number 40 found (*low*)
- Line 159: Missing docstring (*medium*)
- Line 182: Missing docstring (*medium*)
- Line 188: Missing docstring (*medium*)
- Line 195: Missing docstring (*medium*)
- Line 197: Magic number 20 found (*low*)
- Line 210: Missing docstring (*medium*)
- Line 237: Missing docstring (*medium*)
- Line 238: Missing docstring (*medium*)
- Line 241: Missing docstring (*medium*)
- Line 257: Missing docstring (*medium*)
- Line 266: Missing docstring (*medium*)
- Line 268: Missing docstring (*medium*)
- Line 279: Magic number 11 found (*low*)
- Line 280: Magic number 12 found (*low*)
- Line 281: Magic number 13 found (*low*)
- Line 282: Magic number 14 found (*low*)
- Line 283: Magic number 15 found (*low*)
- Line 284: Magic number 16 found (*low*)
- Line 285: Magic number 17 found (*low*)
- Line 286: Magic number 18 found (*low*)
- Line 287: Magic number 19 found (*low*)
- Line 288: Magic number 20 found (*low*)
- Line 289: Magic number 21 found (*low*)
- Line 290: Magic number 22 found (*low*)
- Line 291: Magic number 23 found (*low*)
- Line 292: Magic number 24 found (*low*)
- Line 293: Magic number 25 found (*low*)
- Line 294: Magic number 26 found (*low*)
- Line 296: Missing docstring (*medium*)
- Line 308: Magic number 11 found (*low*)
- Line 309: Magic number 12 found (*low*)
- Line 310: Magic number 13 found (*low*)
- Line 311: Magic number 14 found (*low*)
- Line 312: Magic number 15 found (*low*)

- Line 313: Magic number 16 found (*low*)
- Line 314: Magic number 17 found (*low*)
- Line 315: Magic number 18 found (*low*)
- Line 316: Magic number 19 found (*low*)
- Line 317: Magic number 20 found (*low*)
- Line 318: Magic number 21 found (*low*)
- Line 319: Magic number 22 found (*low*)
- Line 320: Magic number 23 found (*low*)
- Line 321: Magic number 24 found (*low*)
- Line 322: Magic number 25 found (*low*)
- Line 323: Magic number 26 found (*low*)
- Line 324: Magic number 27 found (*low*)
- Line 325: Magic number 28 found (*low*)
- Line 326: Magic number 29 found (*low*)
- Line 327: Magic number 30 found (*low*)
- Line 328: Magic number 31 found (*low*)
- Line 329: Magic number 32 found (*low*)
- Line 330: Magic number 33 found (*low*)
- Line 331: Magic number 34 found (*low*)
- Line 332: Magic number 35 found (*low*)
- Line 337: Missing docstring (*medium*)
- Line 355: Magic number 50 found (*low*)

Suggestions & Fixes

Fix Critical Bugs

There are 2 bugs that need immediate attention

Security Vulnerabilities

Found 5 security issues

Performance Optimizations

Consider fixing 1 performance issues

Code Style Improvements

Found 84 style issues to address

Suggested Fixes

Bug Fix

Original: Line 59: Potential infinite loop detected

Fixed: Fix the logic error

Explanation: Potential infinite loop detected

Bug Fix

Original: Line 161: Division by zero detected

Fixed: Fix the logic error

Explanation: Division by zero detected

Security Fix

Original: Line 15: Hardcoded credential found

Fixed: Fix the security vulnerability

Explanation: Hardcoded credential found

Security Fix

Original: Line 51: Potential Command Injection vulnerability

Fixed: Fix the security vulnerability

Explanation: Potential Command Injection vulnerability